

STATISTICAL ARBITRAGE & BACKTESTING

Mauricio Labadie, PhD

Algorithmic Quant/Trader

Electronic Broker (Investment Bank)

London, UK

Workshop in Quantitative Finance 2016

Facultad de Ciencias, UNAM

Mexico City, 25 May 2016

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1. Introduction

What is a “Quant”

- “Quant” is the short version of “Quantitative Analyst”
- They are the “math gurus” in a financial firm e.g. bank or fund
 - They build the mathematical tools for:
 - Pricing
 - Underwriting
 - Trading
 - Risk management
 - Etc
- Two types of quants:
 - “Probability” quants
 - “Statistics” quants

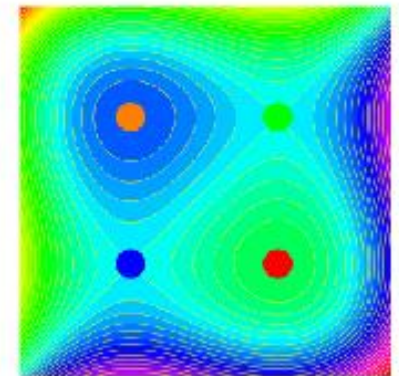
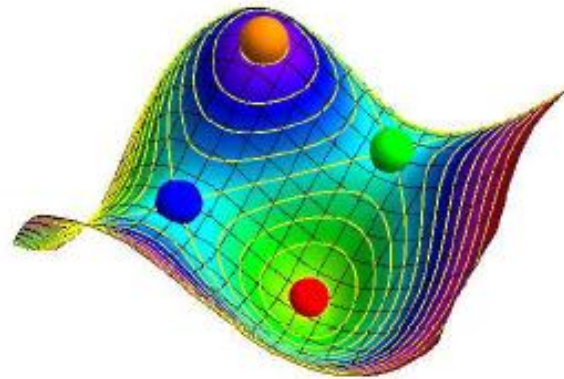


harringtonstarr.com

“Probability” Quant

- They provide the **mathematical models** for the firm
- They need a strong background in Mathematics:
 - Probability
 - Differential equations
 - ODEs
 - PDEs
 - SDEs
 - Numerical methods
 - Optimisation
- And some Economics:
 - Efficient Market Theory
 - Utility functions
 - Cost functions

$$f(x, y) = x^3/3 - x - (y^3/3 - y)$$



math.harvard.edu

“Statistics” Quant

- They provide the **data & analytic tools** for the firm
- They need a strong background in Statistics:
 - Data mining
 - Data crunching
 - Distributions
 - Curve fitting
 - Tests of hypotheses
 - Machine Learning
- And some Economics:
 - Econometrics
 - Macro- and Microeconomics
 - Price-formation processes



harringtonstar.com

2. Statistical Arbitrage

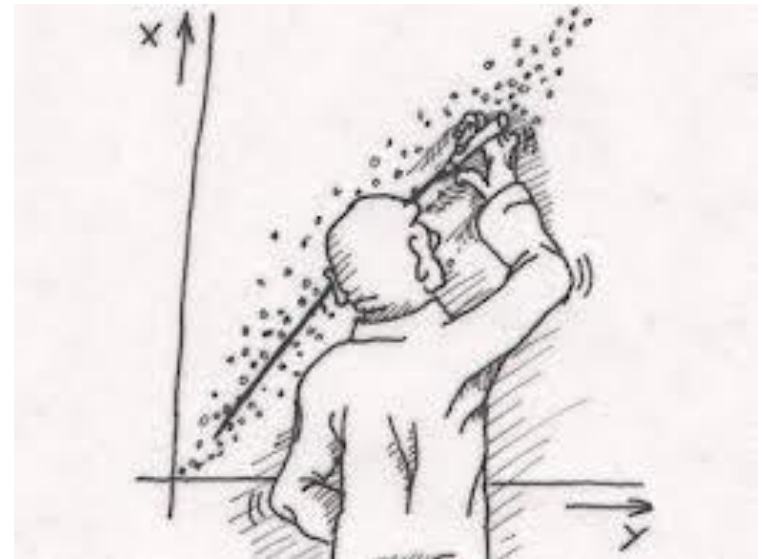
Statistical Arbitrage

- Principle 1
 - There are identifiable patterns in the financial markets
- That means:
 - We can find exploitable trading strategies
- Example:
 - Price trends
 - Volatility trends
 - Volume trends
 - Macro events
 - News



Statistical Arbitrage

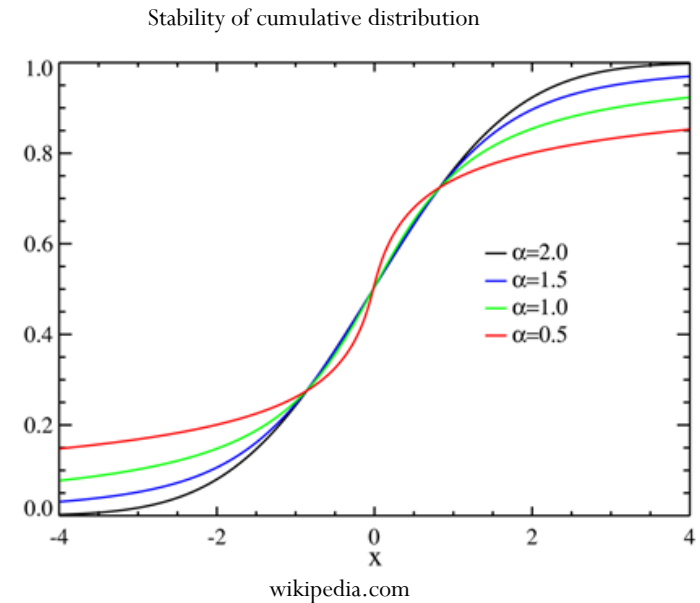
- Principle 2
 - Some identified patterns are statistically robust
- That means:
 - Some patterns are stable under small changes on their input and their parameters
- Example:
 - Parametric models:
 - Model is stable
 - Non-parametric models:
 - Distribution is stable
 - Recurrent behaviour:
 - Volumes and volatility spikes



algo-trades.com

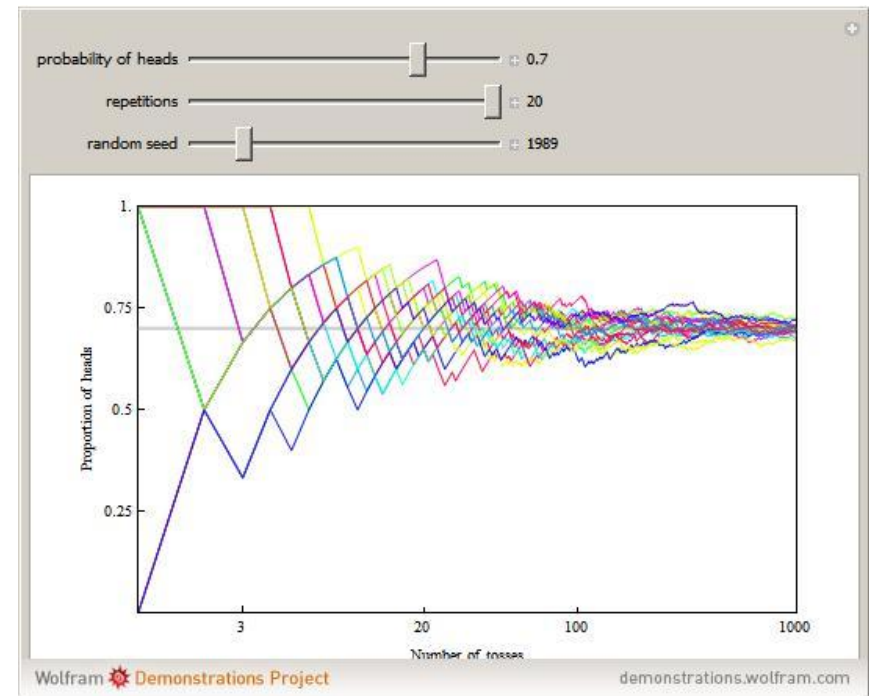
Statistical Arbitrage

- Principle 3
 - For some patterns, past behaviour can on average predict future behaviour
- That means:
 - Some patterns are stable in time
 - At least in the short term
 - Potential need of “periodic recalibration”
- Example:
 - Volume and volatility curves
 - Correlations
- Predicting future prices can be possible but is very subtle



Statistical Arbitrage

- Principle 4
 - A strategy exploiting a robust pattern is profitable on average
- That means:
 - Do not expect to win all the time
 - But you can win in the long run:
 - Law of Large Numbers
 - Central Limit Theorem
- Example:
 - Lottery tickets
 - Insurance premium
 - Option pricing
 - Market-making



wolfram.com

Statistical Arbitrage

- Principle 5
 - A pattern normally changes after some time
- That means:
 - Even if patterns are stable in time, they do not last forever
 - Some patterns can disappear if the market changes
 - Frequent recalibration to determine when a strategy is no longer profitable

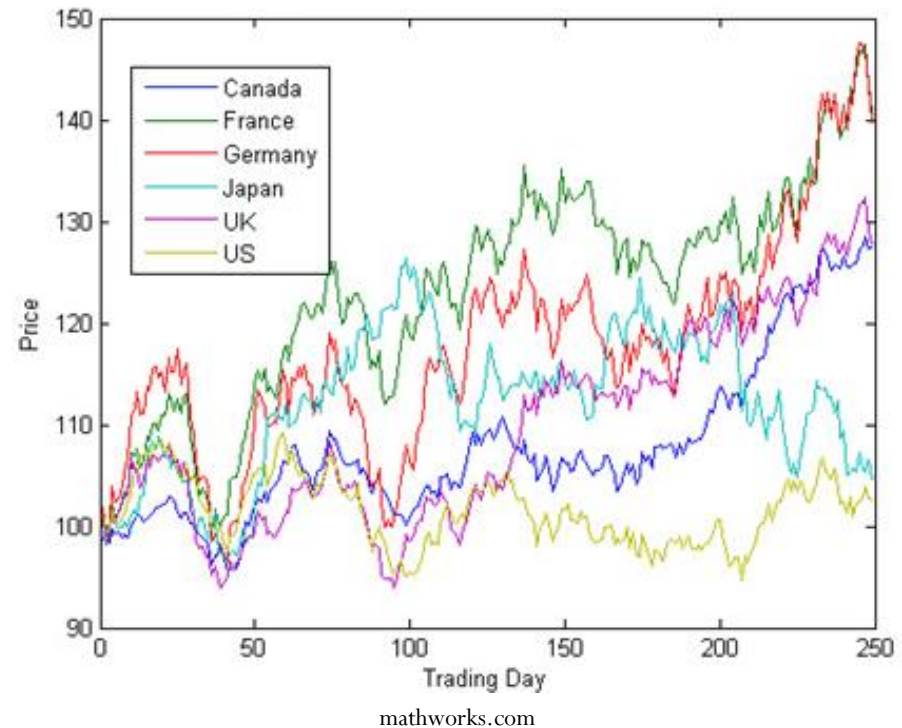
- Example:
 - Correlation strategies:
 - Pair trading
 - Index arb



3. Backtesting

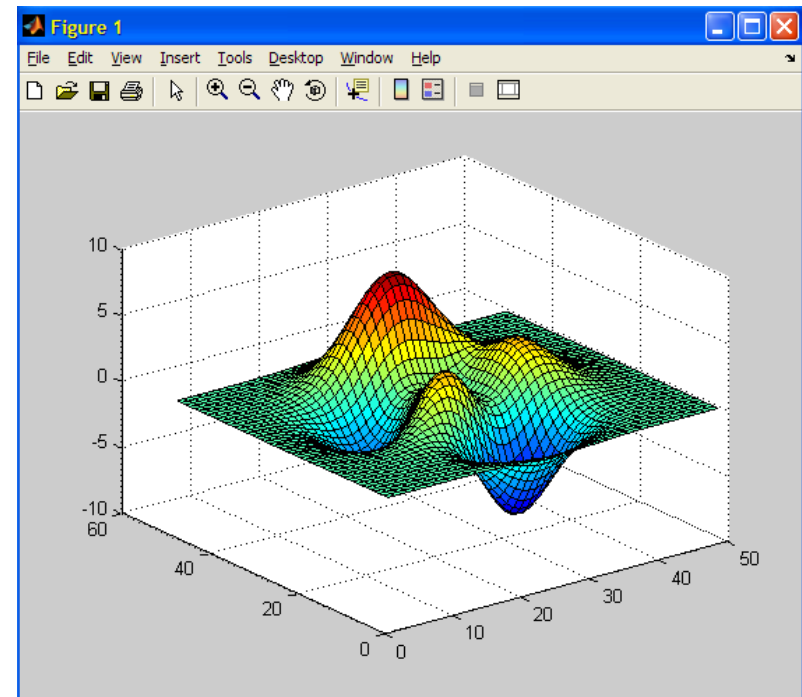
Backtesting

- Stage 1
 - Build prototype of the trading strategy
- What to do:
 - Code the rules of the algorithm
 - Matlab, R or Python
 - Simulate time series:
 - Monte Carlo
 - Use simulations to test:
 - Code
 - Rules
 - Dependence to parameters
 - Get a first glimpse of the distribution



Backtesting

- Stage 2
 - Define the parameters and the “utility function”
- What to do:
 - Define the space of parameters
 - Potentially reduce dimensions
 - Define the optimisation function
 - Utility function
 - Maximise
 - Cost function
 - Minimise



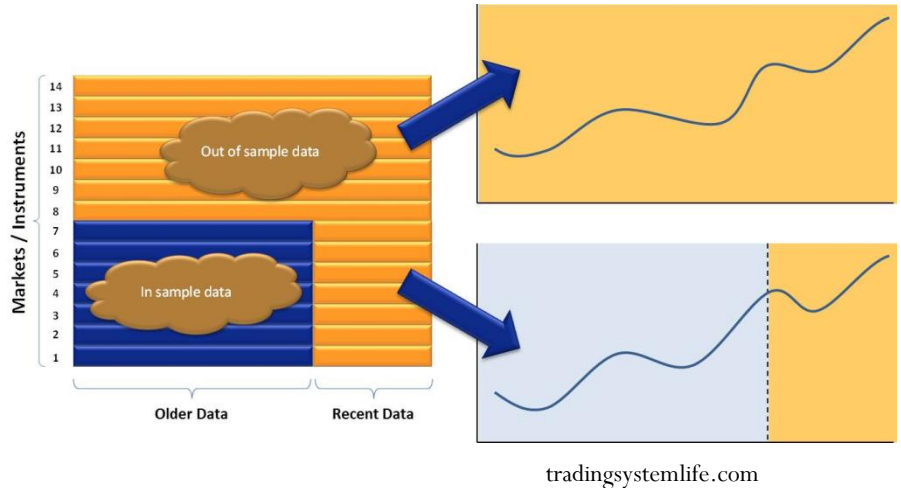
Matlab

Backtesting

- Stage 3
 - Define your “In Sample” and “Out of Sample”

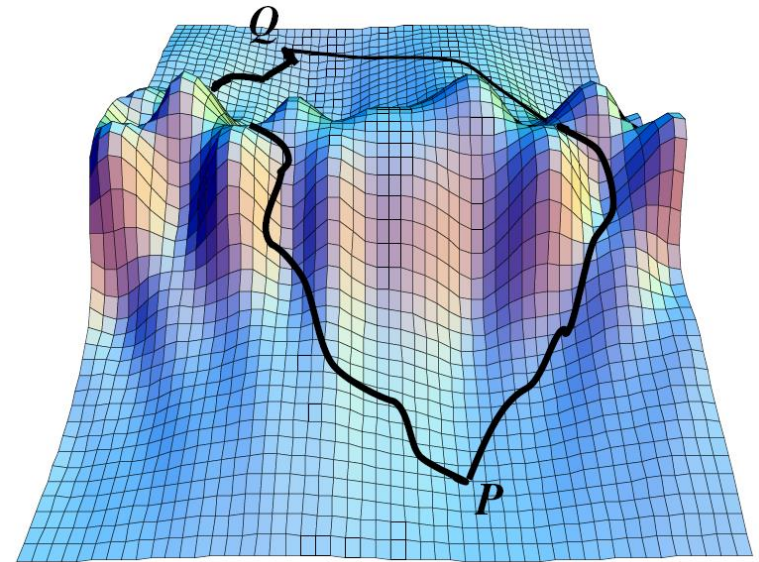
- What to do:

- Get time series of real data:
 - Data needs to be cleaned
- Divide the sample set in two:
 - In Sample:
 - Find optimal parameters
 - Out of Sample:
 - Test statistical robustness of the optimal parameters found In Sample



Backtesting

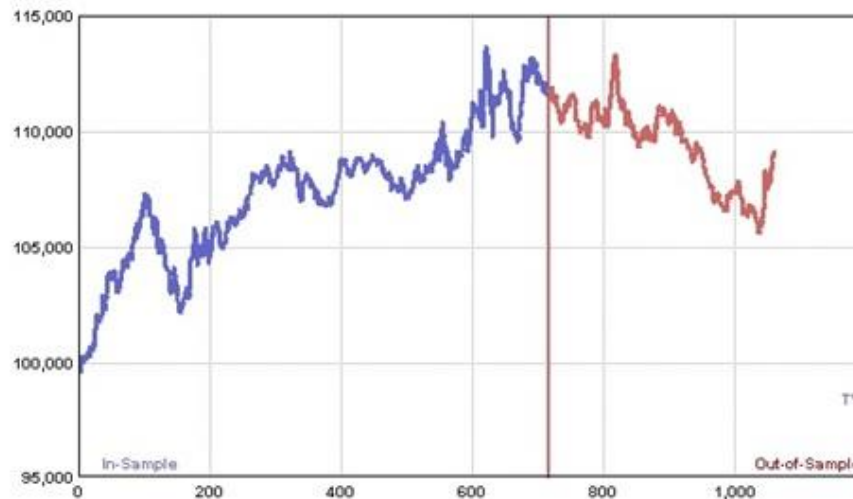
- Stage 4
 - Find the optimal parameters for the utility function
- What to do:
 - Find the best parameters In Sample:
 - By “brute force” if ≤ 2 parameters:
 - Plot the whole utility function
 - Find the global maximum
 - By numerical methods:
 - Gradient methods
 - Beware of
 - Saddle points
 - Local maxima



graphics.ethz.ch/~achapiro

Backtesting

- Stage 5
 - Test the stability of the In-Sample best parameters in the Out of Sample
- What to do:
 - If the pattern is still profitable, keep the strategy
 - Otherwise, discard the strategy and restart from Stage 1

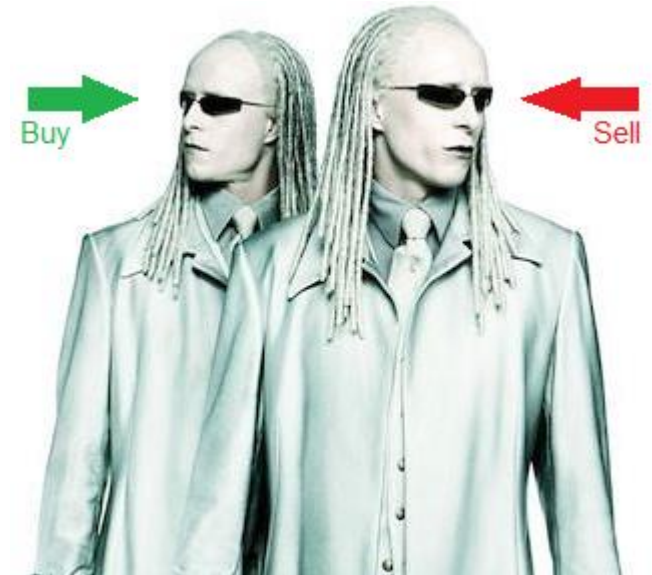


<http://aostrading.cz/>

3. Example: Pairs Trading

Pairs Trading

- Market neutral strategy
 - Buy one stock, sell another
 - Their correlation needs to be strong:
 - Same sector, country, market cap, etc
- Assumptions:
 - Correlation will continue to be strong
 - Any break in correlation is temporary
- The spread is mean reverting
 - Sell spread if it is large:
 - Sell outperforming stock
 - Buy underperforming stock
 - Buy spread if it is small



Film Matrix Reloaded

Pairs Trading

- Pairs trading tries to arbitrage temporary loss of correlation
 - It assumes both stocks will get correlated again
- Danger when the correlation pattern breaks
 - “Stocks fall out of sync”
 - As it happened with equities, rates and oil in the last year



nasdaq.com

Pairs Trading

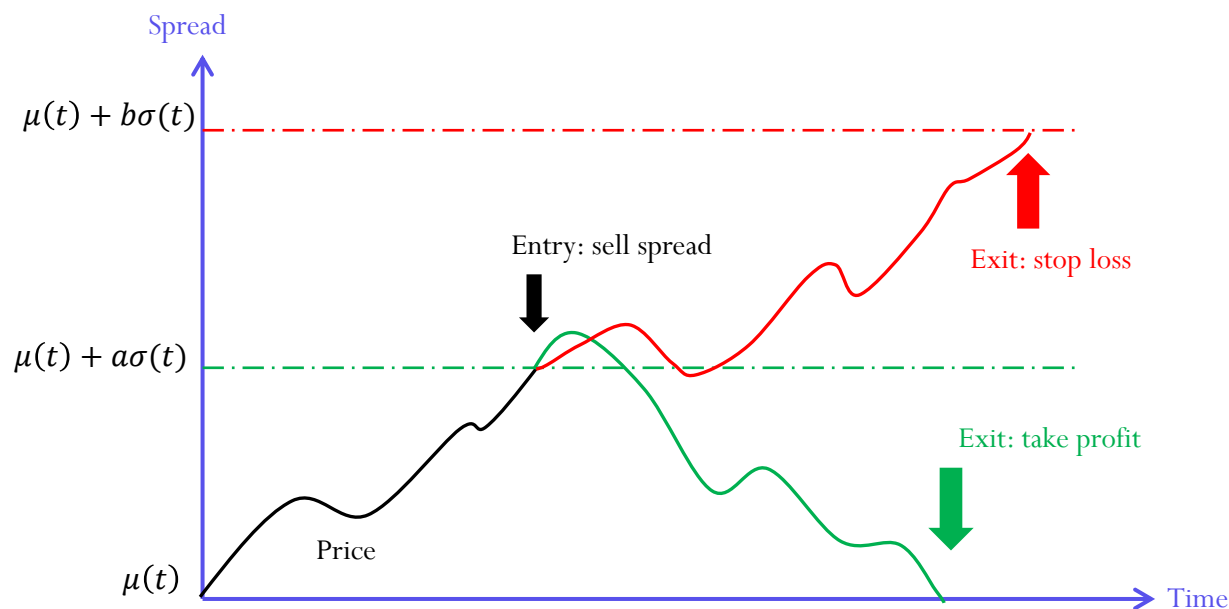
- Too many parameters:
 - Returns:
 - Monthly? Daily?
 - Intraday? Every 5 mins, 10 mins, ..., 1 hour?
 - Entry and Exit signals:
 - Based on rolling mean and volatility of spread
 - But how many points? 10, 20, 30, ...?
 - $1/N$ weights? Exponential weights?
 - Transaction costs and execution:
 - Trading software?
 - Market and broker fees?
 - Market orders? Limit orders?
 - Latency?



elearning.industriallogic.com

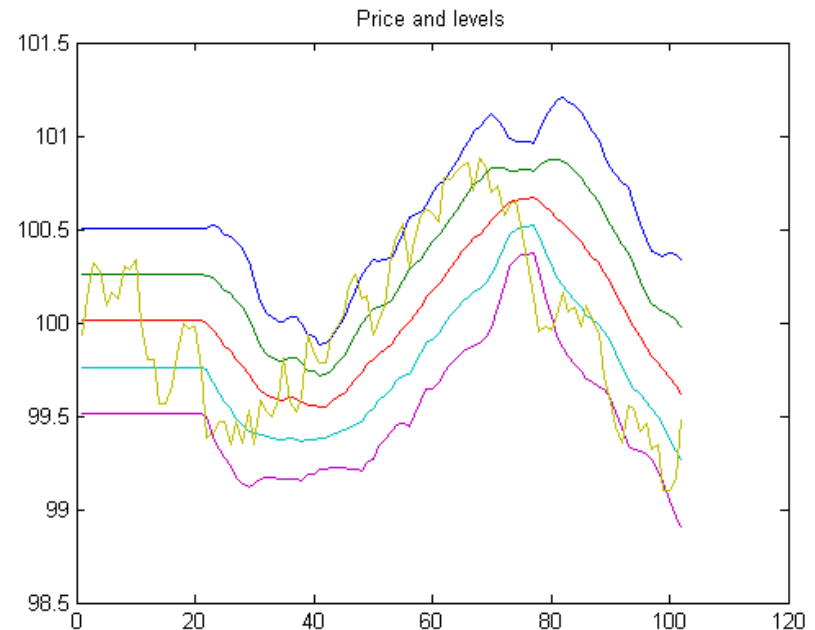
Pairs Trading

- We will use only 2 parameters:
 - The threshold of the entry signal:
 - $\mu(t) \pm a\sigma(t)$
 - The threshold of the stop-loss signal:
 - $\mu(t) \pm b\sigma(t)$, $b > a$



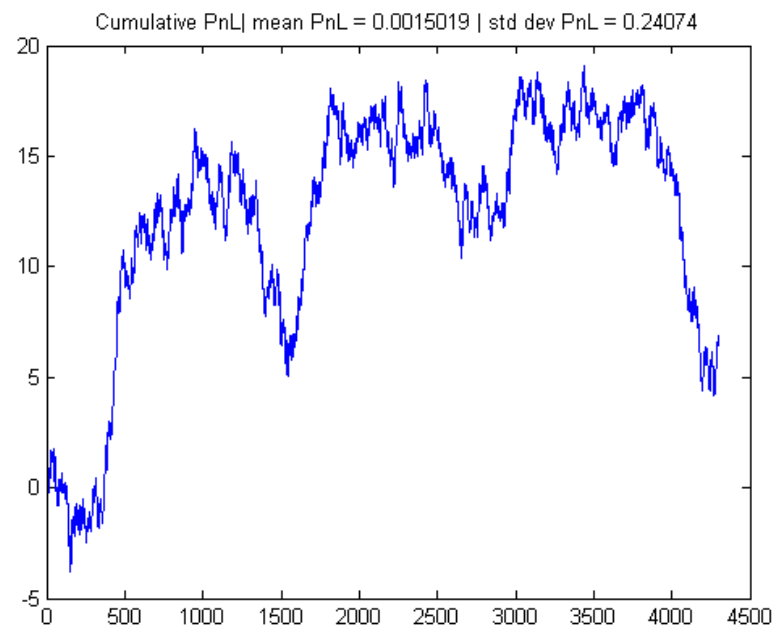
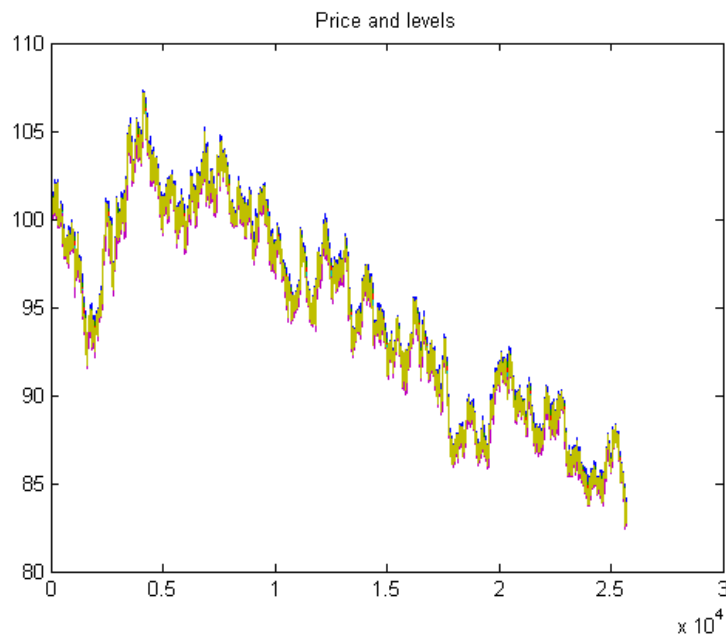
Pairs Trading: Exercise (1)

- Build the algorithm:
 - Create a price simulator
 - Use it to simulate the spread
 - Intraday prices every 5 minutes
 - One day of data
 - Compute the entry and exit levels
 - Rolling mean and volatility
 - In this exercise we used Matlab
 - You can use R, Python, etc



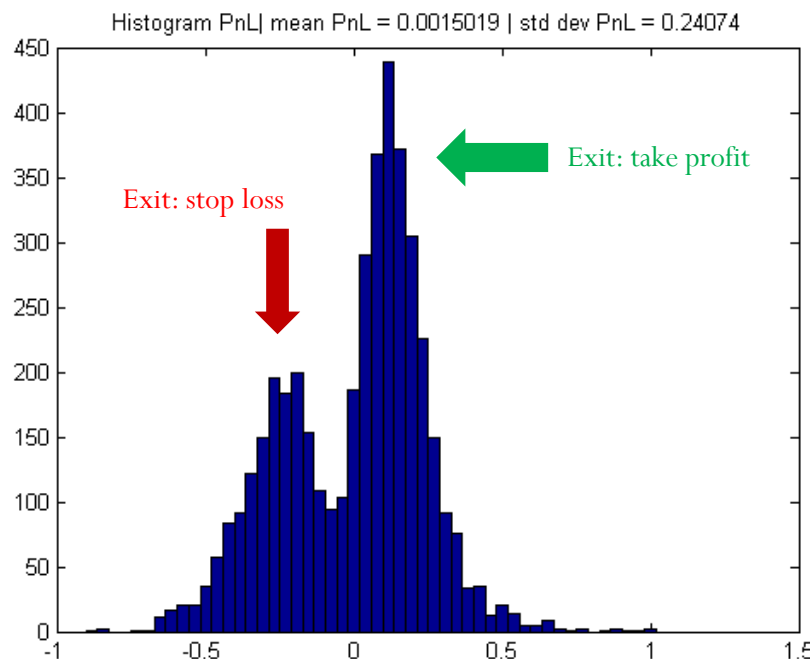
Pairs Trading: Exercise (2)

- Monte Carlo Simulation:
 - Run the algorithm for 252 days
 - Compute:
 - PNL (profit and loss) per transaction
 - Cumulative PNL



Pairs Trading: Exercise (3)

- Monte Carlo Simulation:
 - Build the distribution of PNL
 - Histogram
 - Spread was modelled as a Normal random variable
 - Shall we expect a normal distribution of PNL?



Not very “normal”!

Pairs Trading: Homework (1)

- Gather real time data:
 - Time series (prices) of the 2 legs of the spread:
 - 1 year of data for In Sample
 - e.g. 2015
 - 3-4 months of data for Out of Sample
 - e.g. Jan-Apr 2016
 - From the two time series of prices compute:
 - 5-minute returns
 - Spread as the difference of returns
 - Mean of spread $\mu(t)$
 - Volatility of spread $\sigma(t)$
 - Entry levels $\mu(t) \pm a\sigma(t)$
 - Stop-loss levels $\mu(t) \pm b\sigma(t)$

Pairs Trading: Homework (2)

- Run the algorithm In Sample:
 - Compute PNL
 - Compute the distribution
- Optimisation:
 - Find the values for a and b that:
 - Maximise profit
 - Minimise risk
- Run the algorithm Out of Sample
 - Use the optimal values for a and b found in In Sample
 - Check that the Out of Sample is still profitable

References

- Harris, “*Trading and Exchanges*”, Oxford University Press, 2003
- Barry Johnson, “*Algorithmic Trading & DMA*”, 4Myeloma Press, 2010
- There are many books in Quantitative Trading
- Try some free MOOCs on Machine Learning:
 - Coursera (Stanford University)
 - Udacity (Georgia Tech)



LOTR Movie

Thank you for your
attention



**KEEP
CALM
PRESENTATION IS OVER
ANY
QUESTIONS?**